

DiffAtOnce

Practical spectrometer calibration manual

Instrument preparation, 90° pulse and DOSY scale, step by step

The screenshot displays the DiffAtOnce software interface. At the top, the title bar reads "DiffAtOnce" and "Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results". A language dropdown menu is set to "English". Below the title bar, a status bar indicates "DEMONSTRATION · simulated data · no instrument connection". The main content area features a navigation bar with tabs: "01 Overview", "02 Instrument", "03 Pulse P1", "04 DOSY gradient", and "05 Run & results". The "01 Overview" tab is active, showing a workflow titled "One workflow. Every decision recorded." with the instruction: "Prepare the instrument, calibrate the 90° pulse, then measure a reference gradient ramp. Review signed integrals, attenuation and residuals before applying a result." The workflow consists of four steps: 01 Prepare (Lock, tune/match, shim and thermal equilibration depend on the installed hardware.), 02 Calibrate P1 (A signed nutation series identifies P90. A 22 µs first null gives approximately 11 µs.), 03 Calibrate DOSY (Integrate one reference signal and fit its attenuation at fixed sequence and timings.), and 04 Review & export (Inspect residuals. Save the plan, CSV and report. Copy a reviewed P1 to a new EXPNO.). Below the steps are two buttons: "Configure P1" and "Configure DOSY". A "Scope of this version" section states: "Automates channel-1 P1 and a 1D DOSY reference series. It does not certify every probe/channel or replace service calibration. Hardware-dependent preparation steps are enabled individually. Real-instrument commissioning is still required." At the bottom, a status bar shows "Ready to configure | Jython 2.7 / TopSpin 3 | v1.0" and two buttons: "Load settings" and "Save settings".

Guided instrument preparation, channel-1 P1 and DOSY calibration with a standard. The operator enables capabilities present on the instrument and reviews each action. Software prepares new copies, runs an authorised series, processes it and calculates proposals. It does not measure T1, calibrate other channels, replace service calibration or independently measure true sample temperature. P1 is applied only to a reviewed new copy; global calibrations are not written.

Screenshots show real Java Swing interface components rendered by the application in Demo mode; they are not captures of the spectrometer desktop. Data and instrument are simulated. They do not document acquisition or calibration performed on hardware.

Python / Jython 2.7 + Swing · TopSpin 3 · English / Español

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Recommended reading

Start with Preparation and P1. Then review the DOSY reference, sequence parameters and attenuation. Large figure pages let you zoom into the interface while keeping it readable.

TopSpin launch

```
xpy spectrometer_calibration_gui.py
```

Local demonstration

```
.\run_demo.ps1 -TopSpinRoot C:\Bruker\TopSpin3.8.0
```

01 Getting started: what to calibrate, and in which order

Start by preparing the instrument and calibrating the pulse. Then establish the gradient scale using a diffusion standard. These are different results: P90 describes the radiofrequency pulse duration at a particular power; DOSY calibration relates signal attenuation to diffusion.

1. In Overview, select Español or English and check the connection mode. Use Read current dataset in Instrument and inspect dataset fields and the report folder in P1. Changing the language does not change units or experimental parameters.
2. Follow Instrument → P1 → DOSY → Run & results. Complete preparation first; a software indicator does not replace inspection of the sample, spectral line and instrument controls.
3. Use Demo to learn the workflow. Practise previewing plans, inspecting plots and opening reports before connecting to TopSpin. Always identify whether a result comes from simulation or acquired data.
4. On the real instrument, start from a reviewed template. Choose new EXPNOs to retain every trial and preserve the starting parameters.

An experiment number is an identifier, not a physical property: EXPNO 1000 is the calibration experiment used in this laboratory.

Checking P90 does not require changing the gradient constant. Calibrate and document each operation separately.

Getting started: what to calibrate, and in which order

DiffAtOnce
Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

English

DEMONSTRATION · simulated data · no instrument connection

01 Overview

02 Instrument

03 Pulse P1

04 DOSY gradient

05 Run & results

One workflow. Every decision recorded.

Prepare the instrument, calibrate the 90° pulse, then measure a reference gradient ramp. Review signed integrals, attenuation and residuals before applying a result.

01

Prepare

Lock, tune/match, shim and thermal equilibration depend on the installed hardware.

02

Calibrate P1

A signed nutation series identifies P90. A 22 μ s first null gives approximately 11 μ s.

03

Calibrate DOSY

Integrate one reference signal and fit its attenuation at fixed sequence and timings.

04

Review & export

Inspect residuals. Save the plan, CSV and report. Copy a reviewed P1 to a new EXPNO.

Configure P1

Configure DOSY

Scope of this version

Automates channel-1 P1 and a 1D DOSY reference series. It does not certify every probe/channel or replace service calibration. Hardware-dependent preparation steps are enabled individually. Real-instrument commissioning is still required.

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

Overview: language, context and connection status. Screenshot in Demo mode.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

02 Installation, launch and runtime

The GUI uses Jython 2.7-compatible Python and Java Swing graphical components inside TopSpin 3. It does not require the modern external Python 3 API to control TopSpin. The available local installation is TopSpin 3.8.0; live operation in 3.6.4 must be checked on that installation.

1. Copy the self-contained `dist/spectrometer_calibration_gui.py` file from the package to `<TopSpin>/exp/stan/nmr/py/user/`. Retain the previous version if that name is already in use.
2. Open TopSpin, select a suitable 1D template and enter `xpy spectrometer_calibration_gui.py`. Use Read current dataset in Instrument and check the fields in P1 before preparing destinations.
3. For a standalone demonstration, use `run_demo.ps1`. The launcher uses the local JRE and Jython paths in its configuration; it does not install a runtime or connect to a spectrometer by itself.
4. If Java, Jython or the script cannot be found, inspect the path in the error message. Running an external `python.exe` is not equivalent to opening the GUI in TopSpin: TopCmds calls require that context.

Syntax compatibility and tests with simulated TopCmds check software behaviour, but do not establish instrument communication or the quality of an experimental calibration.

Loading settings restores editable values but clears the active result and sample, hardware and review checks. Repeat these checks for the current session; a saved file cannot establish current instrument conditions.

Installation, launch and runtime

DiffAtOnce
Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

English

DEMONSTRATION · simulated data · no instrument connection

01 Overview

02 Instrument

03 Pulse P1

04 DOSY gradient

05 Run & results

One workflow. Every decision recorded.

Prepare the instrument, calibrate the 90° pulse, then measure a reference gradient ramp. Review signed integrals, attenuation and residuals before applying a result.

01

Prepare

Lock, tune/match, shim and thermal equilibration depend on the installed hardware.

02

Calibrate P1

A signed nutation series identifies P90. A 22 μ s first null gives approximately 11 μ s.

03

Calibrate DOSY

Integrate one reference signal and fit its attenuation at fixed sequence and timings.

04

Review & export

Inspect residuals. Save the plan, CSV and report. Copy a reviewed P1 to a new EXPNO.

Configure P1

Configure DOSY

Scope of this version

Automates channel-1 P1 and a 1D DOSY reference series. It does not certify every probe/channel or replace service calibration. Hardware-dependent preparation steps are enabled individually. Real-instrument commissioning is still required.

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

GUI launch and selected mode; the Demo status remains visible.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

03 Preparing the sample, temperature and signal

Instrument organises separate preparation actions. Create a working copy before running them: temperature commands can update experiment parameters. The original template is retained and the TopSpin view is restored after each action; returning to that view does not propagate parameters from the copy.

1. Insert the sample according to the laboratory procedure, without spinning for DOSY. In Preparation copy EXPNO, enter an unused number such as 1050 and select Create preparation copy. Only parameters are copied from the P1 template, for example 1000; no signal is copied or acquired.
2. Enable only functions available on that instrument. For LOCK, confirm hardware and configured solvent; the solvent must match SOLVENT in the copy. Run LOCK sends lock -acqu. Inspect stability afterwards: a successful command response does not establish stable lock.
3. For Run ATMA, confirm that the probe has compatible ATM hardware. ATMA adjusts tuning and matching using the channels routed in the dataset; it does not calibrate the 90° pulse. If ATM is unavailable, adjust manually using native instrument controls.
4. Enable TopShim configured for this probe only when hardware, probe and setup are compatible. Inspect lock on the correct solvent and confirm that check before Run TOPSHIM. Examine its report and the spectral line shape, not only completion status.
5. For Read controller temperature, enable the controller and command. teget runs in copy 1050 and updates its stored status. A fresh readback belongs to the controller; it establishes neither true sample temperature nor thermal equilibrium. Reading stored TE without teget is not a live sensor query.
6. For Set temperature demand, enter a numerical setpoint in kelvin and explicit minimum/maximum limits appropriate to the probe, sample and controller. taset receives that demand only within the limits. Afterwards query the controller and allow equilibration according to the laboratory procedure.
7. Before the P1 sweep, select an isolated signal near the observation frequency, set RG without clipping the maximum, and fix power, phase and sufficient recovery delay D1. Review the template to be used: select 1050 as the P1 template when its updated parameters are needed, or manually update other working templates while preserving originals.

Changes to TE in 1050 are not automatically transferred to 1000 or DOSY template 10. The submitted demand affects the instrument controller, but each experiment's stored parameters must be reviewed separately.

Scope is guided preparation + channel-1 P1 + standard DOSY. T1 is not determined, nor are other channels, service calibrations or true sample temperature validated.

The Instrument tab scrolls vertically. Scroll to the bottom to see all four operator checks and the Read current dataset button.

Preparing the sample, temperature and signal

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Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

English ▼

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Prepare the measurement conditions first

Enable only capabilities present on your spectrometer. Each command is started separately. Command completion does not prove that tuning, shimming or thermal equilibrium is acceptable.

Preparation copy EXPNO1050

Prepare copy

☐ Lock command available; solvent configured

Run LOCK

☐ Automatic tune/match hardware available

Run ATMA

☐ TopShim configured for this probe

Run TOPSHIM

☐ Lock established and reviewed before TopShim

☐ Temperature controller available

Read temperature

New demand temperature (K)298.15

Probe/controller temperature limits (K) to

Set temperature

TE is saved metadata, not independent sample thermometry. Temperature-controller read/set operations are not issued on original templates.

Operator checks before acquisition

☐ Correct sample, probe, channel and RF power

☐ Temperature equilibrated and reference D(T) reviewed

☐ Template phase, integration region and receiver gain reviewed

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

Real interface: copy 1050, configurable capabilities and temperature controls. Swing components rendered in Demo mode.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

04 P1: configuring a nutation curve

Open the calibration template, for example EXPNO 1000, and confirm PULPROG = zg. This sequence applies P1 directly. The GUI does not automatically interpret stored P1 as a 90° pulse: it may be the value used to find the 180° null.

1. In P1, enter distinct, increasing durations in microseconds. Include the signal rise, positive maximum and points beyond the first inversion. At least nine values are required for fitting.
2. Define a P90 search interval containing the physically expected value. If current P1 is 22 μ s and represents 180°, P90 may be near 11 μ s; do not automatically centre the search on 22.
3. Enter lower and upper integration limits in ppm from the actual spectrum. The window must contain the same complete resonance, without unrelated regions or boundaries outside the spectrum.
4. Select the first new EXPNO. Preview should show one row per duration, its destination and inherited parameters. Review RF power, RG, NS, D1 and PHC0/PHC1 before preparation.

The initial sweep is editable. Review it against the probe's RF limits; the software does not determine suitable power or duration for every hardware configuration.

zg30 is not equivalent to zg: this tool's fit requires the simple zg sequence.

P1: configuring a nutation curve

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Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

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Dataset name

Data directory

Reports directory

PROCNO

Template EXPNO

First new EXPNO

Pulse lengths (µs, comma separated)

P90 search minimum (µs)

P90 search maximum (µs)

Integration lower limit (ppm)

Integration upper limit (ppm)

03 Pulse P1

Keep the RF power, RG, NS, D1 and phase fixed. Use zg and an isolated signal. Include the positive maximum, first null and negative signal. Do not autophase each spectrum.

Your example: template 1000 → first null P180 = 22 µs → P90 = 11 µs. The fitted value remains specific to that RF power and channel.

Preview plan

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

P1 plan: values, P90 interval, ppm region and new destinations. Demo example 22/11 µs.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

05 P1: retaining the sign and recognising the first null

At fixed power, the signal first rises, reaches the 90° maximum, decreases and crosses zero near 180°. That first positive-to-negative crossing gives $P_{90} = P_{180}/2$. The next crossing, at 360°, requires division by four.

1. First acquire a clearly positive signal and set its phase. Retain PHC0/PHC1 throughout the sweep: EFP uses that phase. Do not apply APK independently or switch to magnitude mode.
2. Check that the result includes positive and negative integrals. If all points appear positive, review phase, the P1 interval and processing before accepting a proposal.
3. Compare measured points with the fitted sine and inspect residuals. The model includes amplitude and a baseline offset; RF inhomogeneity, relaxation and off-resonance behaviour may depart from the ideal sine.
4. If the optimum is at a search boundary, widen or correct the interval using physical judgement and inspect the sweep. Do not accept a value solely because its R^2 is high.
5. After reviewing P90, experimentally check for a strong positive signal at that pulse and the same power. Record the result and its conditions before transferring it to another template.

$P_{90} = 11 \mu\text{s}$ and $P_{180} = 22 \mu\text{s}$ are the synthetic demonstration example. They are neither universal values nor a new probe measurement.

P1: retaining the sign and recognising the first null

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01 Overview02 Instrument03 Pulse P104 DOSY gradient05 Run & results

Active calibrationP1

Preview plan

Prepare series

Acquire and fit

Analyze series

100%

Stop after pointStop is cooperative. A running acquisition is allowed to finish; no new point starts afterwards.

Reviewed experiment plan

Point	EXPNO	Parameter	Value
1	1100	P 1	2.0
2	1101	P 1	4.0
3	1102	P 1	6.0
4	1103	P 1	8.0
5	1104	P 1	10.0
6	1105	P 1	12.0
7	1106	P 1	14.0
8	1107	P 1	16.0
9	1108	P 1	18.0
10	1109	P 1	20.0

Execution journal
[processing] 1123 processing
[point_finished] 1123 point_finished
[analyzing] analyzing
[result] result
[finished] completed

☐ I reviewed spectra, fit, RF power and residuals

New destination EXPNO1300

Apply P1 to new EXPNO

Open reports

Fit and residuals
SIMULATED RESULT | P90 11.000 μ s / P180 22.000 μ s

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

Signed curve and P90 proposal; synthetic data in Demo mode.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

2026-10-03 | v1.0 | English

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06 DOSY: choosing the standard, temperature and signal

DOSY calibration requires a standard with known diffusion for its composition, solvent and temperature. Entering a water value from a table is insufficient. Identify the reference-D source and keep it associated with the report.

1. Open a prepared 1D DOSY template with reviewed P90, sequence, gradient shape, P30, D20, D16, RG and phase. This GUI varies GPZ6 within one timing condition.
2. In DOSY, enter the standard identity, solvent and source. Enter Dref in $10^{-9} \text{ m}^2/\text{s}$: typing 1.9 means $1.9 \times 10^{-9} \text{ m}^2/\text{s}$. Enter the reference temperature in kelvin.
3. Set tolerances for TE variation and departure from Tref. The reader compares stored parameters from each acquisition; it does not automatically calculate a temperature-corrected Dref.
4. Choose an isolated standard resonance and a fixed ppm window. Inspect phase and baseline throughout the series, including attenuated points. Integration retains the sign, but the logarithmic DOSY fit requires positive areas.

The retained project comparison uses HDO/D₂O Dref = 1.902 at 298.15 K. Do not assign it to an entire ramp acquired at another temperature.

If sample, composition or temperature changes, review the reference before reusing it.

DOSY: choosing the standard, temperature and signal

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Template EXPNO
10

First new EXPNO
1200

Expected pulse program
stebpgp1s1d

GPZ6 (% maximum, comma separated)
8,12,16,20,24,28,32,36,40,44,48,52,56,60,64,68,72,76,80,84,88,92,96

D reference ($10^{-9} \text{ m}^2/\text{s}$)
1.15

Reference temperature (K)
298.15

Reference / solvent / source
DEMO reference / water / synthetic fixture

Maximum recorded TE span (K)
0.5

Maximum TE-to-reference difference (K)
0.5

Previous b100 (s/m^2 ; 0 = none)
0

Integration lower limit (ppm)
4.6

Integration upper limit (ppm)
5.0

04 DOSY gradient

Use a known D reference at the stated temperature. The entire series uses one pulse sequence, gradient shape, P30 and D20. The fit calibrates b100 empirically; another Δ requires its own calibration or a verified sequence model.

Sequence & timing

P30, D20, D16, D5 and double-stimulated-echo timings are inherited from the reviewed template. Inspect its pulse program. Never substitute the rectangular PGSE equation without checking the sequence.

Template phase, integration region and receiver gain reviewed

Preview plan
Dataset: DEMO_DOSY
P1 template: 1000 | DOSY template: 10

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settingsSave settings

DOSY reference: identity, Dref, temperature and ppm window. Demonstration values.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

07 DOSY: designing the ramp and interpreting its scale

Before a full series, test low, intermediate and high gradients. Aim for attenuation spread across useful points: an almost flat curve provides little information, while signal lost too early leaves few reliable points. Match encoding and scan count to the available signal.

1. Enter GPZ6 as increasing, distinct percentages. The 8-96%, step-4% example gives 23 points. It is configurable; 96% is not a universal recommendation for every probe.
2. Keep Δ , δ and shape fixed within the ramp. For `stebpgp1s1d`, D20 is stored in seconds and each P30 in microseconds; small δ is $2 \times P30$. Check the actual pulse program before using that relationship.
3. The fit plots $\ln(I/I_{\max})$ against $(GPZ6/100)^2$ with a free intercept. The first 8% point does not measure zero-gradient intensity and is not forced to represent zero attenuation.
4. The slope magnitude divided by Dref in SI units gives b100. The relationship $b(g) = b100 \times (GPZ6/100)^2$ belongs to those conditions. After changing Δ or δ , recalculate b with a verified model or calibrate that condition separately.

Use the separate generator for three ramps with configurable Δ . The current calibration GUI analyses one timing condition per series.

DOSY: designing the ramp and interpreting its scale

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Active calibration **DOSY** Preview plan Prepare series **Acquire and fit** Analyze series

100%

Stop after point Stop is cooperative. A running acquisition is allowed to finish; no new point starts afterwards.

Reviewed experiment plan

Point	EXPNO	Parameter	Value
1	1200	GPZ 6	8.0
2	1201	GPZ 6	12.0
3	1202	GPZ 6	16.0
4	1203	GPZ 6	20.0
5	1204	GPZ 6	24.0
6	1205	GPZ 6	28.0
7	1206	GPZ 6	32.0
8	1207	GPZ 6	36.0
9	1208	GPZ 6	40.0
10	1209	GPZ 6	44.0

Execution journal

[processing] 1222 processing
[point_finished] 1222 point_finished
[analyzing] analyzing
[result] result
[finished] completed

☐ I reviewed spectra, fit, RF power and residuals

New destination EXPNO 1300

Apply P1 to new EXPNO

Open reports

Fit and residuals
SIMULATED RESULT | $b_{100} = 2e+09 \text{ s/m}^2$

$R^2 = 1.000000$
Use a known D reference at the stated temperature. The entire series uses one pulse sequence, gradient shape, P30 and D20. The fit calibrates b_{100} empirically; another Δ requires its own calibration or a verified sequence model.

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settingsSave settings

Plan and result of a DOSY gradient ramp with fixed timing. Demo mode.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

08 Preview, preparation, acquisition and stopping

Review the plan before starting an action. Exporting or displaying a plan lets you inspect it; preparation creates parameter-only experiments; acquisition performs measurements on the connected instrument. Demo mode reproduces the workflow with simulated data and must remain identified as such.

1. For initial preparation, destinations must be unused and exclude the template. Prepare new experiments copies parameters through wraparam and checks every point. Afterwards, the same GUI session can acquire those destinations if it recognises the prepared plan and its files remain unchanged, without a previous FID.
2. When choosing acquisition, review and confirm the start. The executor waits for ZG, checks its completion status, runs EFP with inherited phase and reads the processed spectrum before continuing.
3. During the series, follow the active EXPNO and Run & results log. Do not manually change the dataset or parameters that the sweep must retain.
4. Stop requests termination after the current point. It does not immediately abort radiofrequency activity in an acquisition already running. To interrupt that point, use native TopSpin controls and the instrument procedure.
5. After an acquisition stop, retain partial results and the log. A partially acquired series is neither resumed nor overwritten: inspect what happened and choose new EXPNOs. Analyze existing series reviews previously acquired data without running ZG; specify the matching EXPNOs and sweep values.

An existing FID does not establish successful acquisition completion. An ambiguous command status stops the workflow even when a file has been written.

Preview, preparation, acquisition and stopping

DiffAtOnce
Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

English

DEMONSTRATION · simulated data · no instrument connection

01 Overview02 Instrument03 Pulse P104 DOSY gradient05 Run & results

Active calibrationDOSY

Preview plan

Prepare series

Acquire and fit

Analyze series

47%

Stop after pointStop is cooperative. A running acquisition is allowed to finish; no new point starts afterwards.

Reviewed experiment plan

Point	EXPNO	Parameter	Value
1	1600	GPZ 6	8.0
2	1601	GPZ 6	12.0
3	1602	GPZ 6	16.0
4	1603	GPZ 6	20.0
5	1604	GPZ 6	24.0
6	1605	GPZ 6	28.0
7	1606	GPZ 6	32.0
8	1607	GPZ 6	36.0
9	1608	GPZ 6	40.0

Fit and residuals

Run or analyze a series to see the measured curve and its residuals.

Execution journal

[stopped] Stopped between points at operator request
Partial results retained; use NEW EXPNO for a new run.
C:/DiffAtOnce/python/calibration_gui/demo_evidence/gui_gradient_DEMO_20261003T214030Z_6d9a683c\executio
[finished] stopped

☐ I reviewed spectra, fit, RF power and residuals

New destination EXPNO1300

Apply P1 to new EXPNO

Open reports

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

Execution and log. Stop acts after the current point; demonstration without hardware.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

09 Reviewing proposals and applying P1 to a new copy

Calculating a proposal and applying it are different operations. Review the curve, residuals and conditions before copying P90. This GUI's apply-P1 action prepares a new parameter-only EXPNO; it preserves the original and does not change a global instrument calibration.

1. Check sample identity, probe, nucleus and RF power. A proposal obtained at one power must not be transferred to another without checking the pulse relationship.
2. Select the destination template and a new EXPNO, review proposed P1 and confirm the copy. Then open the destination and verify P1 together with inherited power.
3. Inspect related pulses in the DOSY program. Do not assume every 180° pulse is an independent parameter that can be updated automatically: some sequences define it from P1.
4. For the gradient, the usual result is b100. Conversion to G requires a verified sequence and shape model; if unavailable, retain the empirical scale without inventing a physical gradient.
5. The separate correction $G_{\text{new}} = G_{\text{previous}} \times \sqrt{D_{\text{measured}}/D_{\text{ref}}}$ requires D_{measured} to have been calculated using that exact G_{previous} . Review units before recording a constant in Bruker controls.

The GUI does not write gradient_calib or the shared TBO profile. Reviewing a proposal is not independent experimental verification.

Reviewing proposals and applying P1 to a new copy

DiffAtOnce
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01 Overview02 Instrument03 Pulse P104 DOSY gradient05 Run & results

Active calibrationP1

Preview plan

Prepare series

Acquire and fit

Analyze series

100%

Stop after pointStop is cooperative. A running acquisition is allowed to finish; no new point starts afterwards.

Reviewed experiment plan

Point	EXPNO	Parameter	Value
1	1100	P 1	2.0
2	1101	P 1	4.0
3	1102	P 1	6.0
4	1103	P 1	8.0
5	1104	P 1	10.0
6	1105	P 1	12.0
7	1106	P 1	14.0
8	1107	P 1	16.0
9	1108	P 1	18.0
10	1109	P 1	20.0

Execution journal
[processing] 1123 processing
[point_finished] 1123 point_finished
[analyzing] analyzing
[result] result
[finished] completed

☒ I reviewed spectra, fit, RF power and residuals

New destination EXPNO1300

Apply P1 to new EXPNO

Open reports

Fit and residuals
SIMULATED RESULT | P90 11.000 μ s / P180 22.000 μ s

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

P1 and new-destination review before copying parameters. Demo mode.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

10 Two complete examples to practise

The following values are for walking through Demo mode. Pulse durations and gradients for a real measurement must be chosen for the sample and instrument limits. Use the examples to check the reasoning and units.

1. P1 example: a template displays $P1 = 22 \mu\text{s}$. Enter nine or more values covering the rise, the maximum near $11 \mu\text{s}$ and inversion near $22 \mu\text{s}$; include later negative points. Set a search interval containing $11 \mu\text{s}$.
2. Review the plan, run Demo and observe the sign change. The synthetic model should propose $P90 \approx 11 \mu\text{s}$ and $P180 \approx 22 \mu\text{s}$. Inspect residuals, then practise copying parameters to a new destination.
3. DOSY Demo example: retain one D20 and P30 condition and use 8-96% in 4% steps. The synthetic reference uses $D_{\text{ref}} = 1.15$ and $T_{\text{ref}} = 298.15 \text{ K}$. Its known b_{100} is $2 \times 10^9 \text{ s/m}^2$. These values define simulated data; they are not a new measurement of water or the instrument.
4. The expected synthetic slope is -2.3 against squared relative gradient: $b_{100} = 2.3 / (1.15 \times 10^{-9}) = 2 \times 10^9 \text{ s/m}^2$. At 50%, $b = 5 \times 10^8 \text{ s/m}^2$. Check that the report recovers this scale and retains the Demo identity, reference and conditions.

These numerical results describe mathematical examples. A Demo screenshot may use other configured values: compare with its plan and report rather than a memorised expected number.

Two complete examples to practise

DiffAtOnce
Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

English

DEMONSTRATION · simulated data · no instrument connection

01 Overview02 Instrument03 Pulse P104 DOSY gradient05 Run & results

Active calibration DOSY Preview plan Prepare series Acquire and fit Analyze series

100%

Stop after point Stop is cooperative. A running acquisition is allowed to finish; no new point starts afterwards.

Reviewed experiment plan

Point	EXPNO	Parameter	Value
1	1200	GPZ 6	8.0
2	1201	GPZ 6	12.0
3	1202	GPZ 6	16.0
4	1203	GPZ 6	20.0
5	1204	GPZ 6	24.0
6	1205	GPZ 6	28.0
7	1206	GPZ 6	32.0
8	1207	GPZ 6	36.0
9	1208	GPZ 6	40.0
10	1209	GPZ 6	44.0

Execution journal

[processing] 1222 processing
[point_finished] 1222 point_finished
[analyzing] analyzing
[result] result
[finished] completed

☐ I reviewed spectra, fit, RF power and residuals

New destination EXPNO

Apply P1 to new EXPNO Open reports

Fit and residuals
SIMULATED RESULT | $b_{100} = 2e+09 \text{ s/m}^2$

$R^2 = 1.000000$
Use a known D reference at the stated temperature. The entire series uses one pulse sequence, gradient shape, P30 and D20. The fit calibrates b_{100} empirically; another Δ requires its own calibration or a verified sequence model.

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings Save settings

Teaching example: check the curve, units and result conditions.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

11 Resolving errors while retaining traceability

Read the last log event and affected EXPNO before repeating. An error may concern preparation, communication, processing or fitting. Correcting its cause is more useful than changing thresholds until a number appears.

1. Existing destination: choose unused EXPNOs. Do not delete previous experiments to pass the check; they may contain complete or partial data explaining an interrupted run.
2. P1 lacks negative signal or the optimum lies at a boundary: check fixed phase, duration range, ppm window and recovery. Extend the sweep only within reviewed instrument limits.
3. DOSY has zero or negative area: inspect phase, baseline, overlap and noise. Do not replace areas by absolute values; that transformation can turn a signal problem into apparent attenuation.
4. Incompatible temperature or parameters: identify what changed. Stabilise the sample, separate conditions or supply the appropriate reference. Increasing a tolerance needs justification and must remain recorded.
5. Ambiguous ZG/EFP, empty FID or incomplete 1r: stop the sequence, inspect TopSpin and retain execution.json. File existence alone does not confirm success.
6. Dataset, RG or phase changed during the series: inspect the responsible intervention or configuration. Then start a new consistent series with unused destinations.

The first on-instrument test must also check return states from that specific TopSpin version. Demo behaviour does not establish those states.

Resolving errors while retaining traceability

DiffAtOnce

Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

English

DEMONSTRATION · simulated data · no instrument connection

01 Overview

02 Instrument

03 Pulse P1

04 DOSY gradient

05 Run & results

Active calibration

P1

Preview plan

Prepare series

Acquire and fit

Analyze series

0%

Stop after point

Stop is cooperative. A running acquisition is allowed to finish; no new point starts afterwards.

Reviewed experiment plan

Point	EXPNO	Parameter	Value
1	1100	P 1	2.0
2	1101	P 1	4.0
3	1102	P 1	6.0
4	1103	P 1	8.0
5	1104	P 1	10.0
6	1105	P 1	12.0
7	1106	P 1	14.0
8	1107	P 1	16.0
9	1108	P 1	18.0

Fit and residuals

Run or analyze a series to see the measured curve and its residuals.

Execution journal

Destination EXPNO already exists. Choose a new range: 1100, 1101, 1102, 1103, 1104, 1105, 1106, 1107, 1108, 1109, 1110
SYNTHETIC DEMO: outputs are not experimental measurements
Destinations already exist: preparation/acquisition will be blocked

☐ I reviewed spectra, fit, RF power and residuals

New destination EXPNO

1300

Apply P1 to new EXPNO

Open reports

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings

Save settings

Example of a controlled stop with cause and EXPNO recorded; Demo mode.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

12 Saving, checking and communicating the result

Keep the plan, execution log and results together. They allow reconstruction of what was attempted, which points completed and which data entered the fit. An attractive plot or high R^2 does not replace this information.

1. Use Open report folder. Inspect plan.json, workflow.json and, for preparation or acquisition, execution.json. Distinguish prepared, acquired, processed, analysed and stopped. A prepared series is not yet a measurement.
2. The GUI saves P1 or DOSY results in result.json and points, predictions and residuals in curve.csv. DOSY also produces informe.html. The earlier console assistant uses different filenames; do not confuse them when locating this GUI's output.
3. Check sample, EXPNO/PROCNO, ppm region, power, sequence, timings, gradient, TE and Dref source. Hashes identify the files read; they do not independently establish scientific quality.
4. Verify P90 with a checking acquisition. Verify the DOSY scale with a new standard measurement under documented conditions. Refitting the same data to the chosen Dref is not an independent test.
5. When sharing results, state Demo or experiment, software version and verification scope. Backend tests with simulated TopCmds are not real acquisition and do not validate TopSpin 3.6.4 on this computer.

The DOSY slope standard error covers regression, not all uncertainty in the standard, temperature, processing or model. The sine-fit R^2 also does not quantify physical P90 uncertainty.

Saving, checking and communicating the result

DiffAtOnce
Spectrometer Calibration / From instrument preparation to reviewed P1 and DOSY results

English

DEMONSTRATION · simulated data · no instrument connection

01 Overview 02 Instrument 03 Pulse P1 04 DOSY gradient 05 Run & results

Active calibration **DOSY**
Preview plan
Prepare series
Acquire and fit
Analyze series

100%

Stop after point
Stop is cooperative. A running acquisition is allowed to finish; no new point starts afterwards.

Reviewed experiment plan

Point	EXPNO	Parameter	Value
1	1200	GPZ 6	8.0
2	1201	GPZ 6	12.0
3	1202	GPZ 6	16.0
4	1203	GPZ 6	20.0
5	1204	GPZ 6	24.0
6	1205	GPZ 6	28.0
7	1206	GPZ 6	32.0
8	1207	GPZ 6	36.0
9	1208	GPZ 6	40.0
10	1209	GPZ 6	44.0

Execution journal

```
[processing] 1222 processing
[point_finished] 1222 point_finished
[analyzing] analyzing
[result] result
[finished] completed
```

☐ I reviewed spectra, fit, RF power and residuals

New destination EXPNO
Apply P1 to new EXPNO
Open reports

Fit and residuals
SIMULATED RESULT | $b_{100} = 2e+09 \text{ s/m}^2$

$R^2 = 1.000000$
Use a known D reference at the stated temperature. The entire series uses one pulse sequence, gradient shape, P30 and D20. The fit calibrates b_{100} empirically; another Δ requires its own calibration or a verified sequence model.

Ready to configure | Jython 2.7 / TopSpin 3 | v1.0

Load settings
Save settings

Access to the report folder from the results. Demo provenance is retained in the files.

Render of the real Swing interface in Demo mode; synthetic data and no experimental acquisition.

Installation and final check

The ZIP includes one self-contained TopSpin script: `dist/spectrometer_calibration_gui.py`. Copy it to `<TopSpin>/exp/stan/nmr/py/user` using the laboratory procedure, retaining earlier versions. Then run `xpy spectrometer_calibration_gui.py`.

You can also launch the workspace copy using `xpy` followed by its absolute path, if your installation accepts paths in `xpy`. Do not switch to an external Python 3 interpreter expecting it to import `TopCmds`.

`run_demo.ps1` uses Java and Jython from an existing installation. Set `-TopSpinRoot` for TopSpin 3.6.4. The launcher locates the installed `jython*.jar`; it downloads and installs no dependencies.

The GUI dispatches work to a `CmdThread` through `EXEC_PYSCRIPT`. If the dispatcher cannot reach the GUI instance, it stops. Verify this communication using Read current dataset before any acquisition.

Software tests: 51 cases passed in Jython 2.7.2 (18 controller and 33 instrument-preparation cases); CPython passed 41 and skipped the 10 Java-specific cases. Tests used a simulated API. Figures were rendered by the real Swing interface while reading and fitting synthetic Bruker files. $P90 = 11 \mu\text{s}$ and $b100 = 2 \times 10^9 \text{ s/m}^2$.

In the source GUI before this public edition, launch and a CURDATA read through a callback on `CmdThread` were verified inside TopSpin 3.8.0, with no hardware commands or parameter writes. These checks do not validate real acquisition or TopSpin 3.6.4. Commissioning must confirm communication, ZG/EFP return status and each enabled preparation command, template preservation, fixed phase, RF limits and experimental quality.

Public edition: views were rerendered from the actual GUI using public display paths. Synthetic data and fits are unchanged. Native receipts redact private names and paths. The rebuilt public bundle has not been validated on an instrument.

Sources and reference files

Acquisition, nutation and execution journal backend

`python/topspin_console_v3/dosy_autorun.py`

TopSpin calibration v3 usage and limits

`docs/es/bruker-paso-a-paso.md`

Existing v3 backend verification record

`docs/es/verificacion.md`

Processed Bruker reader and empirical DOSY calibration

`python/topspin_console/dosy_calibration.py`

Native DOSY assistant documentation

`docs/es/bruker-paso-a-paso.md`

Manual P90 calibration guidance and original evidence

`docs/en/bruker-step-by-step.md`

Shared TBO experimental calibration

Bruker TopSpin Python Interface

<https://www.bruker.com/en/products-and-solutions/mr/nmr-software/topspin/topspin-python-interface.html>

R. Kerssebaum, Bruker DOSY and Diffusion by NMR

https://2210pc.chem.uic.edu/nmr/downloads/dosy_ts13.pdf

Guided instrument setup and working-copy controls

`python/calibration_gui/instrument_setup.py`

Calibration GUI workflow backend

`python/calibration_gui/calibration_workflow.py`

GUI demonstration verification record

`python/calibration_gui/GUI_VERIFICATION.json`

Additional instrument documentation

Bruker python.pdf pp. 7-8, 20-21; acquisition-reference.pdf pp. 114-116, 199-201; Atma.pdf pp. 19-20; topshim.pdf pp. 5-9. Local documentation inspected from TopSpin 3.8.0. Confirm commands and hardware on the target system.